

Aman Tiwari

B.Tech. III Year, Electronics & Communication Engineering
Indian Institute of Technology Roorkee

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Profile

AI/ML, Large Language Models, Computer Vision, NLP, Data Science, Deep Learning, Data Structures and Algorithms
AIR-797 JEE Advanced 2023; **AIR-640** JEE Mains 2023; **4th Place** Flipkart Grid 6.0 Robotics Track;
KVPY AIR-372; **IOQM Merit Awardee** (2021, 2022)

Education

Indian Institute of Technology, Roorkee B.Tech., Electronics & Communication Engineering	2023 – Present CGPA: 8.556
ALPINE ACADEMY, Indore (CBSE) Intermediate (Class XII)	2023 95.20%
ALPINE ACADEMY, Indore (CBSE) Matriculation (Class X)	2021 96.20%

Research Publications

Urban Cell Planning: AI-Driven Convex Optimization Approach with Field Measurement IEEE ANTS 2025

- Led data-driven comparative cellular coverage analysis for 2 operators using MATLAB on **20k+** samples/operator.
- Trained ML model with **96% accuracy** on RSS data, generating an AI-derived path loss equation matching **3GPP TR**.
- Leveraged CVX (MOSEK) optimisation, determining new tower location to elevate minimum signal strength > -90dBm.

Projects

- Autonomous Multi-Agent Conference Planner — IIT Roorkee** Apr 2026 – Present
- Unified 5-agent **LangGraph** orchestration with QdrantDB RAG for 8K+ event records to automate conference planning.
 - Programmed agent pipelines with shared routing and prompt templates, reducing latency by 25%.
 - Used **Tavily** web search, Groq API, robust 3-stage JSON parser, and **Streamlit** dashboard for multi-agent planning.
- Real-Time Sports Chatbot using RAG (Pathway AI) — IIT Roorkee** Apr 2025
- Developed a real-time sports chatbot with RAG and LLMs for live commentary and human-like responses during events.
 - Automated data ingestion using **Celery task and Crontab**, achieving 24/7 scraping and indexing of real-time data.
 - Devised 3 structured pipelines for Cricket, F1 and Football to extract ball2ball events, race results, and match summaries.
- Smart Vision Quality Control System — MaRS IIT Roorkee** Oct 2024 – Dec 2024
- Engineered a real-time product inspection system using **LLM** and visual data in logistics.
 - Processed **500+** product images, achieving **92% OCR** accuracy and **85%** precision in freshness detection.
 - Segmented using **SAM2** for barcode and multilingual text recognition, automating diverse packaging formats.
- Traffic Light Controller Using Image Sensing — MaRS IIT Roorkee** Aug 2024 – Nov 2024
- Built **YOLOv8** system, detecting vehicles in real-time with **92%** accuracy for flexible traffic signals.
 - Applied OpenCV and RMS clustering for traffic-density estimation across 5K+ vehicle detections
 - Custom-trained model with CVAT annotations to prioritise emergency vehicles (**~1.5k samples**) via signal control.
- Sound-based Drone Fault Classification with Multi-Task Learning — MaRS IIT Roorkee** Mar 2024 – Apr 2024
- Architected a **multi-task ML pipeline** using sound recordings to classify drone type, flight direction, and fault presence.
 - Processed and analysed **65k+** drone audio samples, extracting MFCCs and frequency domain features for classification.
 - Achieved **90%+ accuracy** across all 3 tasks using Random Forest and CNN-based classifiers.

Technical Skills

- **Languages:** Python, C++, SQL, HTML, CSS, JavaScript, Bootstrap, ROS2
- **Frameworks/Tools:** LangChain, LangGraph, Tavily, QdrantDB, Groq API, Streamlit, AWS, TensorFlow, PyTorch, Keras, OpenCV, NumPy, Selenium, BeautifulSoup, Requests

Positions of Responsibility

- AI/ML Head — Team Robocon IITR** Jun 2025 – Present
- Managed AI/ML division, building low-latency (<30%) object detection models for robotics.
 - Guided juniors and integrated ML pipelines with hardware.
- Technical Head — GambitoR, IITR** Jun 2025 – Present
- Organised outreach drives, including 3-day IITR campus visit for 50+ students.
 - Conducted motivational sessions for 200+ JEE aspirants.
 - Managed outreach website and provided counselling support for admissions.